

# UNLOX MINOR PROJECT-5

## Local AI Study Assistant Using Ollama and Streamlit

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### 1. Project Title

Local AI Study Assistant Using Ollama and Streamlit

### 2. Introduction

The Local AI Study Assistant is a beginner-friendly question-answering application developed using Python, Ollama, Gemma 3 1B, and Streamlit. The application allows users to enter questions through a web interface and receive AI-generated responses from a locally running language model.

The main objective of this project is to understand how a local Large Language Model (LLM) can be integrated with a Python-based user interface.

### 3. What is Ollama?

Ollama is a tool that enables users to download and run Large Language Models locally on their computers. It manages model execution and provides an interface through which applications can communicate with the model.

In this project, Ollama is used to run the Gemma 3 1B model locally without requiring an external cloud-based AI API.

### 4. What is Streamlit?

Streamlit is an open-source Python framework used to create interactive web applications easily. It allows developers to build user interfaces directly using Python without requiring extensive HTML, CSS, or JavaScript knowledge.

In this project, Streamlit is used to create the chatbot interface, input area, response display, sidebar, and chat history.

### 5. Ollama Model Used

The model used in this project is:

**Gemma 3:4B**

This lightweight model was selected because it is suitable for local execution, provides useful responses for general study questions, and is appropriate for a beginner-level AI application.

## 6. Working of the Application

The application follows the following process:

1. The user enters a question in the Streamlit chat interface.
2. Python receives the user's prompt.
3. The prompt is sent to Ollama using the Ollama Python library.
4. Ollama processes the prompt using the Gemma 3 1B model.
5. The generated response is returned to the Python application.
6. Streamlit displays the response to the user.
7. Conversation history is maintained using Streamlit session state.

### Basic Workflow:

User Input → Streamlit → Python → Ollama → Gemma 3 1B → AI Response → Streamlit

## 7. Features

- Local AI question answering
- Interactive chat interface
- Conversation history
- Clear chat functionality
- No external cloud API required
- Simple and beginner-friendly implementation

## 8. Prompts Tested

The application was tested using the following five prompts:

1. Explain polymorphism in Java in simple words with an example.
2. What is the difference between a stack and a queue?
3. Explain the difference between a list and a tuple in Python.
4. What is the difference between Artificial Intelligence and Generative AI?
5. Give me a short revision note on database normalization.

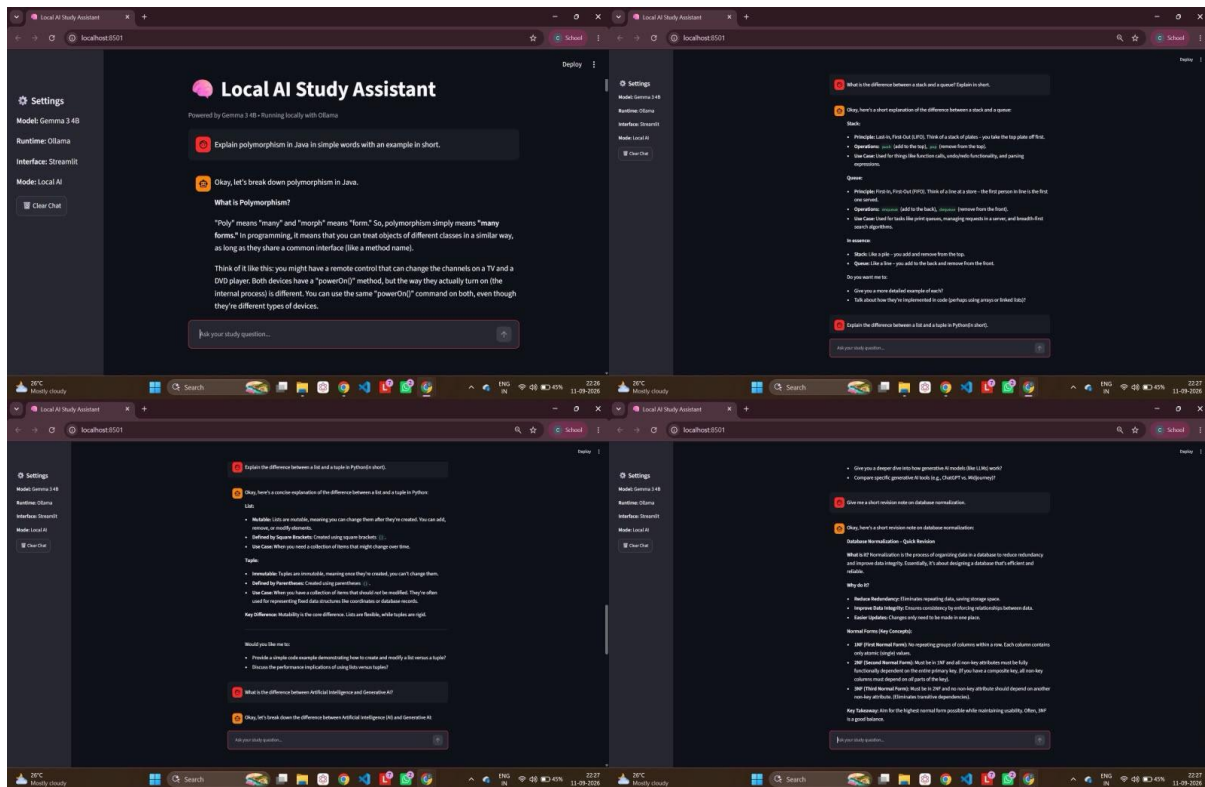
All five prompts were successfully processed and responses were displayed in the Streamlit application.

## 9. Screenshots

The screenshots demonstrate:

- Main application interface
- AI-generated response
- Conversation history

- Additional successful prompt response



## 10. Learning Outcomes

Through this project, I learned how to:

- Install and run a local AI model using Ollama
- Connect Python with Ollama
- Build an interactive web interface using Streamlit
- Use the Ollama Python library
- Maintain conversation history using session state
- Test and document a local AI application

## 11. Conclusion

This project successfully demonstrates the integration of a local language model with a Python-based web interface. The Local AI Study Assistant provides a simple and effective way to ask questions and receive AI-generated responses while running the model locally. The project helped develop practical knowledge of local LLMs, Python integration, and Streamlit application development.